

Composition Determines Diversity

Categorical Fingerprints of Genetic Algorithms

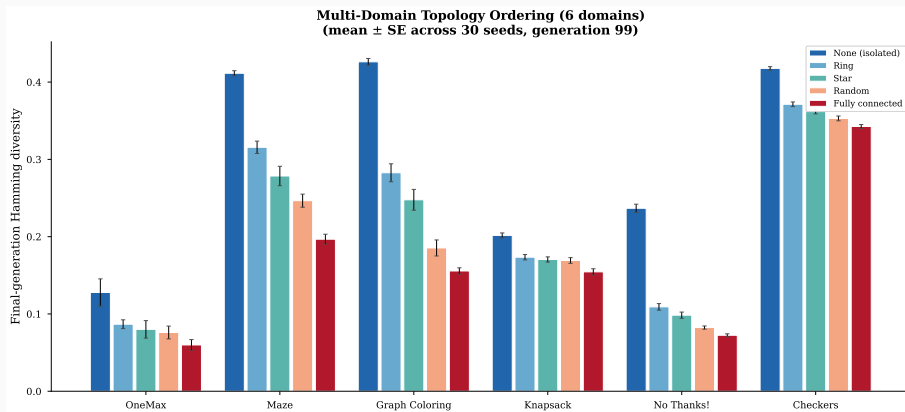
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The Claim



$$W = 1.0$$

Perfect concordance

$$p = 0.00008$$

Not noise

$$23.9\times$$

Topology $\gg$ domain

Why This Matters

Migration topology explains

23.9× more variance

in diversity than domain choice.

The thing you set once and forget
matters more than everything you tune.

Every GA Is a Pipeline

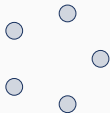
Every GA composes the same operators:

`select → crossover → mutate → evaluate`

(Kleisli composition)

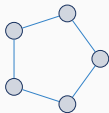
Five Topologies

None



$$\lambda_2 = 0$$

Ring



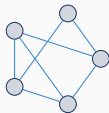
$$\lambda_2 = 1.38$$

Star



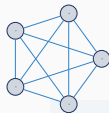
$$\lambda_2 = 1.00$$

Random



$$\lambda_2 \approx 2.5$$

FC



$$\lambda_2 = 5.0$$

More connected $\rightarrow$ faster mixing $\rightarrow$ lower diversity

The Experiment

- 6 domains: OneMax, maze, graph coloring, knapsack, No Thanks!, checkers
- 5 topologies $\times$ 30 seeds
- 5 islands $\times$ 16 individuals
- 100 generations
- **900 runs total**

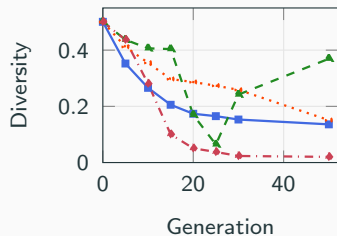
Diversity = normalized pairwise
Hamming distance

Chosen to span:

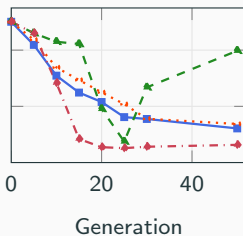
- Trivial $\rightarrow$ NP-hard
- Fixed $\rightarrow$ co-evolutionary
- Binary $\rightarrow$ real-valued

Diversity Fingerprints

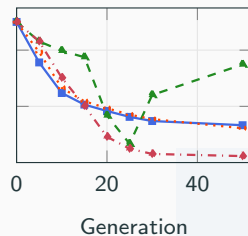
(a) Maze



(b) Graph Coloring



(c) Knapsack



— Flat - - Hourglass Island - . - Adaptive

Same operators. 18x spread.

Three Things To Do Monday

1. **Choose topology FIRST**

Before tuning operators — it dominates everything else

2. **Monitor λ_2 , not diversity directly**

Cheap early-warning signal, computed from topology alone

3. **Add cycle-closing edges**, not migration rate

Changes structure, not just coupling strength

Thank You

Questions?

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github.com/RaggedR/lyra-paper

*“How you wire your islands matters
more than what you evolve”*